R6 10 kΩ
R7 15 kΩ
5 V to the pH board (from the 5V pin)
3.3 V
GND
GPIO4: 5 V × 0.4 = 2.0 V
ESP32-C3 SuperMini
0 1 2 3 4 3.3 G 5V
ESP32-C3
FH4
RST
BOOT
21 20 10 9 8 7 6 5
GND VCC SCL SDA
SSD1306
128 × 64
OLED SSD1306, address 0x3C
SDA
SCL
VDD
GND
SCL
SDA
ADDR
ALRT
A0
A1
A2
A3
ADS1115
A: simulator voltage, taken BEFORE the 50 MΩ resistor
push button
50 MΩ
A3: board current (V on Rsh)
S: up to 5 V, the ADS gets half
R3 10 kΩ
R4 10 kΩ
board GND
Rsh 10 Ω
shunt
S +5V GND
to pH board (JST 3-pin, P2)
6.8 kΩ
LR44 battery, 1.5 V
LR44
1
2
polarity switch
(DPDT, ON-OFF-ON)
3 2 1
trimmer 10 kΩ
(leg 2 = wiper)
to BNC center
BNC female
shield
center
Rt 9.73 MΩ, test only
(known currents, Section 7.3)
reference: pot leg 1, BNC shield, input A1

Key points

A1: to the pot leg that goes to the BNC shield, not to the leg with the 6.8 kΩ resistor.
A0: only BEFORE the 50 MΩ resistor.
Never wire the BNC center to the ADS.
Button pressed: 50 MΩ shorted (calibration).
Button released: leak measurement.
Power the ADS1115 and OLED from 3.3 V.
Feed S to the ADS only through R3 and R4.
Board current: I = V(A3) / Rsh.
S in firmware: 2 × V(A2) − I · Rsh.
A1 has no divider: A is valid only with a board.
Cable to P2 is crossed: connect by function.
Dot = connection. Arc = wires only cross.

- 3.3 V
- 5 V to the pH board
- SDA
- after 50 MΩ, to BNC center
- S, pH board output
- pH board GND, through Rsh
- GND (calibrator ground)
- SCL
- A, wiper (before 50 MΩ)
- reference: pot leg 1, shield, A1
- battery, switch, 6.8 kΩ